

Sangbeom Park

AI RESEARCH ENGINEER

Seoul, Republic of Korea

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Work Experience

ATLO

CTO, FULL-TIME

Seoul, S.Korea

Jun. 2024 – Apr. 2025

- Developed and deployed an AI social robot service using LLMs and multimodal pipelines

NAVER Cloud

MACHINE LEARNING ENGINEER, INTERN

Gyeonggi, S.Korea

Mar. 2024 – Jun. 2024

- Developed NewsBot, a personalized news delivery system powered by HyperCLOVA X

NAVER LABS

MACHINE LEARNING ENGINEER, INTERN

Gyeonggi, S.Korea

Mar. 2023 – Sep. 2023

- Implemented unsupervised skill learning to develop safe and diverse quadruped locomotion behaviors (Mentor: Taeyoon Lee)

Education

Korea University

MASTER OF SCIENCE IN ARTIFICIAL INTELLIGENCE

Seoul, S.Korea

Sep. 2021 – Feb. 2025

- Thesis: Learning Diverse Manipulation Skills for Effective Semi-Autonomous Teleoperation
- Cumulative GPA: 4.42/4.5 | Credit: 40
- Supervisor: Sungjoon Choi

Dongguk University

BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONICS ENGINEERING & MINOR IN INTELLIGENT ROBOTICS ENGINEERING

Seoul, S.Korea

Mar. 2015 – Aug. 2021

- EE GPA: 4.09/4.5 | IR GPA: 4.43/4.5 | Credit: 149
- Leave of absence for military service in KATUSA: Nov. 2016 – Aug. 2018

Publications

Quality-Diversity based Semi-Autonomous Teleoperation using Reinforcement Learning

Neural Networks

SANGBEOM PARK, TAERIM YOON, JOONHYUNG LEE, SUNGHYUN PARK, AND SUNGJOON CHOI

Nov. 2024

- [Project Page](#) | [Paper](#) | [Code](#) | [Poster](#)
- Presented on Physical Human-Robot Interaction at *ICRA 2024 Workshop*
- Listed in COMPUTER SCIENCE, ARTIFICIAL INTELLIGENCE & NEUROSCIENCES (*Impact Factor: 7.8, JCR Top 10%*)

Visual Preference Inference: An Image Sequence-Based Preference Reasoning in Tabletop Object Manipulation

IROS

JOONHYUNG LEE, SANGBEOM PARK, YONGIN KWON, JEMIN LEE, SUNGJOON CHOI

Oct. 2024

- [Project Page](#) | [Paper](#) | [Code](#) | [Video](#) | [Poster](#)
- Presented on Vision-Language Models for Navigation and Manipulation at *ICRA 2024 Workshop*

SPOTS: Stable Placement of Objects with Reasoning in Semi-Autonomous Teleoperation Systems

ICRA

JOONHYUNG LEE, SANGBEOM PARK, JEONGEUN PARK, KYUNGJAE LEE, AND SUNGJOON CHOI

May. 2024

- [Project Page](#) | [Paper](#) | [Code](#) | [Video](#)

CLARA: Classifying and Disambiguating User Commands for Reliable Interactive Robotic Agents

JEONGEUN PARK, SEUNGWON LIM, JOONHYUNG LEE, **SANGBEOM PARK**, MINSUK CHANG, YOUNGJAE YU, AND SUNGJOON CHOI

RA-L

Dec. 2023

- [Project Page](#) | [Paper](#) | [Code](#) | [Video](#)

Semi-Autonomous Teleoperation via Learning Non-Prehensile Manipulation Skills

SANGBEOM PARK, YOONBYUNG CHAI, SUNGHYUN PARK, JEONGEUN PARK, KYUNGJAE LEE AND SUNGJOON CHOI

ICRA

May. 2022

- [Paper](#) | [Code](#) | [Video](#)

Learning Latent Prior for Rapid Adaptation of Legged Robots to Unexpected Amputation

SUNGHYUN PARK, YOONBYUNG CHAI, SEUNGYUP KA, HYEONSEONG KIM, **SANGBEOM PARK**, KEVIN GIM, JOOHYUNG KIM, AND SUNGJOON CHOI

Under Review

2025

Projects

Interactive Multimodal Agent for Emotion-Aware Dialogue

AI RESEARCH ENGINEER & CTO

ATLO

Jun. 2024 - Apr. 2025

- Designed a multimodal interaction pipeline using asynchronous processing.
- Implemented a multimodal RAG pipeline using LangChain and LLMs with prompt engineering.
- Launched and maintained a multimodal AI agent service via a web application *Atto* and iOS app *Ditto*.
- Raised 110 million KRW in seed funding at a 1.1 billion KRW valuation from Strong Ventures.
- Accumulated 54,000+ conversation sessions over four months.

AI News Curation System

MACHINE LEARNING ENGINEER & INTERN

Naver Cloud

Mar. 2024 - Jun. 2024

- Built backend infrastructure using AWS and Docker; designed a MySQL schema for news articles.
- Collected labeled dataset with LLMs and fine-tuned HCX model for news classification.
- Used embedding similarity and reranking to filter duplicates and prioritize high-quality articles.
- Deployed as a chatbot service within a corporate app, regularly delivering curated news to 900+ employees.

Safe and Low-Noise Locomotion Skill Learning

MACHINE LEARNING ENGINEER & INTERN

NAVER LABS

Mar. 2023 - Sep. 2023

- Implemented a Safe Critic Network to suppress unsafe or noisy walking behaviors.
- Applied Hierarchical RL to learn diverse gaits using safety-aware low-level policies.
- Enhanced sim-to-real transfer via domain randomization in Isaac Gym.
- Built a Python-C++ control pipeline with LCM to execute model actions on robot motors in real time.
- Trained stable walking skills with 24% less noise, validated in both sim and real-world settings.

Automation Maintenance using Mobile Manipulator

RESEARCHER

SAMSUNG

Mar. 2021 - Sep. 2023

- Built simulation environments using Unity and Gazebo to test autonomous maintenance tasks such as navigation, grasping.
- Developed navigation and manipulation skills of mobile manipulator for factory automation system.

Presentations

Physical Human-Robot Interaction

CONFERENCE PRESENTATIONS

Yokohama, Japan

May, 2024

- Poster presentation at *IEEE International Conference on Robotics and Automation (ICRA)*

Learning X in Physics-based Simulators

TECH TALK

Seoul, S.Korea

Mar. 2024

- Invited talk at *CINAMON, AI Research Group*

Shared-Autonomy for Robotics

TECH TALK

Gyunggi, S.Korea

Feb. 2023

- Invited talk at *NAVER LABS, Robot Dynamics & Control Group*

Telerobotics

CONFERENCE PRESENTATIONS

Philadelphia, U.S.A

May, 2022

- Oral presentation at *IEEE International Conference on Robotics and Automation (ICRA)*

Patent

2023 **10-2023-0125345**, Semi-Autonomous Teleoperation via Learning Manipulation

Korea

Skills

Programming	Python, C++, C#, Java
Physics Simulation	Isaac Sim, MuJoCo, Unity, Gazebo
Realworld Robots	<u>UR5e</u> , <u>Mini-Cheetah</u> , <u>Franka Panda</u> , <u>Pioneer 3-DX</u> , <u>Xycar</u>
DevOps	Amazon ECS, Docker
Languages	Korean (Native), English (Fluent)